1 Computing Definite Integrals

1.1 Computing definite integrals

To compute the definite integral[1] of function $f(x)$ between limits a and b, which we denote as

$$\int_a^b f(x)\ dx \tag{1}$$

we have two options:

- Compute the Riemann Sum 1.1.1.
- Use the Fundamental Theorem of Calculus 1.1.2.

1.1.1 Compute the Riemann Sum

To compute the definite integral 1 using Riemann sums, we need to compute the Riemann sum explicitly. However, computing a Riemann sum explicitly is not always feasible: for a general continuous function, you can define the integral as the limit of Riemann sums, but evaluating that limit analytically may be difficult. Numerical Riemann sums instead give approximations.

1.1.2 Use the Fundamental Theorem of Calculus

To compute the definite integral 1 using the Fundamental Theorem of Calculus,

1. First, we compute the family of **antiderivatives**[2] of function $f(x)$, which we call the "*indefinite integral of* $f(x)$" and denote as

$$\int f(x)\ dx. \tag{2}$$

2. Then, we use the Fundamental Theorem of Calculus:
   (a) Evaluate one of the **antiderivatives**[2] in the "*indefinite integral of* $f(x)$", 2, at input a.
   (b) Evaluate one of the **antiderivatives**[2] in the "*indefinite integral of* $f(x)$", 2, at input b.
   (c) Finally, by the Fundamental Theorem of Calculus, the definite integral 1 is equal to; the result of step 2b minus the result of step 2a.

[1] A Definite Integral is the continuous analog of a sum, and is used to calculate areas, volumes, and their generalizations: A definite integral computes the signed area of the region in the plane that is bounded by the graph of a given function between two points in the real line.

[2] An "antiderivative of function $f(x)$" is a differentiable function of x whose derivative is $f(x)$.

2 Example

$$\int_0^1 -2e^{-x^2} x \, dx \tag{3}$$

2.1 Using the Fundamental Theorem of Calculus

1.

$$\begin{aligned} f(x) &= -2e^{-x^2} x \\ &= e^{-x^2} (-2x). \end{aligned} \tag{4}$$

Notice that if we define $h(z) = -z^2$, then

$$f(x) = e^{-x^2} \cdot \frac{d}{dx} [h(x)] \tag{5}$$

Now, if we define $g(z) = e^z$, then

$$\begin{aligned} f(x) &= \frac{d}{dh(x)} [g(h(x))] \cdot \frac{d}{dx} [h(x)] \\ &= \frac{d}{dx} [g(h(x))] \end{aligned} \tag{6}$$

or equivalently

$$f(x) = \frac{d}{dx} \left[e^{-x^2} \right] \tag{7}$$

That is,

$$e^{-x^2} \text{ is an antiderivative of } f(x). \tag{8}$$

Finally, notice that

$$\begin{aligned} f(x) &= \frac{d}{dx} \left[e^{-x^2} \right] + 0 \\ &= \frac{d}{dx} \left[e^{-x^2} \right] + \frac{d}{dx} [C], \qquad \text{for all } C \in \mathbb{R}. \\ &= \frac{d}{dx} \left[e^{-x^2} + C \right], \qquad \text{for all } C \in \mathbb{R}. \end{aligned} \tag{9}$$

Therefore,

$$\int f(x) \, dx = e^{-x^2} + C, \qquad \text{for } C \in \mathbb{R}. \tag{10}$$

However, we denote it as $F(x)$.

2. (a) $F(0) = e^{-(1)^2} + C$
   (b) $F(1) = e^{-(0)^2} + C$

(c)

$$\begin{aligned}\int_0^1 -2e^{-x^2} x \, dx &= F(1) - F(0) \\ &= \left[e^{-(1)^2} + C\right] - \left[e^{-(0)^2} + C\right] \\ &= e^{-1} - e^0\end{aligned} \tag{11}$$

3 Example 2

$$\int_0^1 \ln(1+x)\ dx \tag{12}$$

3.1 Using the Fundamental Theorem of Calculus

1. We would like to get rid of all the differentiation operators ($\frac{d}{d[x]}$ or $\square'$)

$$\begin{aligned} f'(x) &= \ln(1+x) \\ &= \ln(1+x)\cdot(1) \end{aligned} \tag{13}$$

Notice that if we define $h(x) = 1+x$, then

$$f'(x) = \ln(h(x))\cdot h'(x) \tag{14}$$

Now, let us assume that

$$\ln(h(x)) = g'(h(x)), \tag{15}$$

for some $g(z)$. Then

$$\begin{aligned} f'(x) &= g'(h(x))\cdot h'(x) \\ &= \frac{d}{d[h(x)]}[g(h(x))]\cdot\frac{d}{d[x]}[h(x)] \\ &= \frac{d}{d[x]}[g(h(x))] \end{aligned} \tag{16}$$

Now, we can find function $g(z)$, by noting that

$$\begin{aligned} g'(h(x)) &= \ln(h(x)) \\ &= \ln(h(x))\cdot(1) \end{aligned} \tag{17}$$

Notice that if we define $u(h(x)) = \ln(h(x))$ and $v(h(x)) = h(x)$, then

$$\begin{aligned} \frac{d}{d[h(x)]}[u(h(x))\cdot v(h(x))] &= u(h(x))\cdot\frac{d}{d[h(x)]}[v(h(x))] + \frac{d}{d[h(x)]}[u(h(x))]\cdot v(h(x)) \\ &= g'(h(x)) + \frac{d}{d[h(x)]}[u(h(x))]\cdot v(h(x)) \\ &= \frac{d}{d[h(x)]}[g(h(x))] + \frac{d}{d[h(x)]}[u(h(x))]\cdot v(h(x)) \end{aligned} \tag{18}$$

That is,

$$\begin{aligned} \frac{d}{d[h(x)]}[g(h(x))] &= \frac{d}{d[h(x)]}[u(h(x))\cdot v(h(x))] - \frac{d}{d[h(x)]}[u(h(x))]\cdot v(h(x)) \\ &= \frac{d}{d[h(x)]}[u(h(x))\cdot v(h(x))] - \frac{1}{h(x)}\cdot h(x) \\ &= \frac{d}{d[h(x)]}[u(h(x))\cdot v(h(x))] - 1 \end{aligned} \tag{19}$$

Or equivalently,

$$\begin{aligned} 1 &= \frac{d}{d\left[h\left(x\right)\right]}\left[u\left(h\left(x\right)\right)\cdot v\left(h\left(x\right)\right)\right]-\frac{d}{d\left[h\left(x\right)\right]}\left[g\left(h\left(x\right)\right)\right] \\ &= \frac{d}{d\left[h\left(x\right)\right]}\left[u\left(h\left(x\right)\right)\cdot v\left(h\left(x\right)\right)-g\left(h\left(x\right)\right)\right] \end{aligned} \tag{20}$$

That is,

$$h\left(x\right)=u\left(h\left(x\right)\right)\cdot v\left(h\left(x\right)\right)-g\left(h\left(x\right)\right) \tag{21}$$

That is,

$$\begin{aligned} g\left(h\left(x\right)\right) &= u\left(h\left(x\right)\right)\cdot v\left(h\left(x\right)\right)-h\left(x\right) \\ &= \ln\left(h\left(x\right)\right)\cdot h\left(x\right)-h\left(x\right) \\ &= \ln\left(1+x\right)\cdot\left(1+x\right)-\left(1+x\right) \end{aligned} \tag{22}$$

Therefore

$$\frac{d}{d\left[x\right]}\left[f\left(x\right)\right]=\frac{d}{d\left[x\right]}\left[g\left(h\left(x\right)\right)\right] \qquad \Longrightarrow \qquad f\left(x\right)=g\left(h\left(x\right)\right)=\ln\left(1+x\right)\cdot\left(1+x\right)-\left(1+x\right) \tag{23}$$

4 Integration by substitution

We try to find an appropiate $h(x)$, such that

$$\begin{aligned}\frac{d}{d[x]}[f_0(x)] &= \frac{d}{d[h(x)]}[f_1(h(x))] \cdot \frac{d}{d[x]}[h(x)] \\ &= \frac{d}{d[x]}[f_1(h(x))]\end{aligned} \tag{24}$$

If we find such $f_1(h(x))$, then it follows that

$$f_0(x) = f_1(h(x)) \tag{25}$$

5 Integration by parts

We try to find appropiate $u(x)$ and $v'(x)$, such that

$$\begin{aligned}\frac{d}{d[x]}[f_0(x)] &= u(x) \cdot \frac{d}{d[x]}[v(x)] \\ &= \frac{d}{d[x]}[u(x) \cdot v(x)] - \frac{d}{d[x]}[u(x)] \cdot v(x)\end{aligned} \tag{26}$$

That is,

$$\begin{aligned}\frac{d}{d[x]}[u(x)] \cdot v(x) &= \frac{d}{d[x]}[u(x) \cdot v(x)] - \frac{d}{d[x]}[f_0(x)] \\ &= \frac{d}{d[x]}[u(x) \cdot v(x) - f_0(x)]\end{aligned} \tag{27}$$

If we know $v(x)$, then we can try to find $f_1(x)$ such that

$$\frac{d}{d[x]}[f_1(x)] = \frac{d}{d[x]}[u(x)] \cdot v(x) \tag{28}$$

If we find such $f_1(x)$, then it follows that

$$f_1(x) = u(x) \cdot v(x) - f_0(x) \tag{29}$$

Or equivalently,

$$f_0(x) = u(x) \cdot v(x) - f_1(x) \tag{30}$$